Figures in support of paper Why communicating minimum drawdown elevations for Lake Powell and Lake Mead can mitigate risks of volatile Colorado River flow.

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This report was generated on: November 4 , 2026

Overview

This is an R Markdown document. This document generates Figures 1, 2, and 4 in support of the paper "Why communicating minimum drawdown elevations for Lake Powell and Lake Mead can mitigate risks of volatile Colorado River flow."

Figures:

1. Lake Powell storage over time in relation to critical volumes.
2. Lake Mead storage over time in relation to critical volumes.
3. Recent Lake Powell storage (black) with example minimum drawdown and catastrophic elevations (red dashed).

Additional Figures:

AA. Lake Powell, Lake Mead, and Combined Storage. BB. Lake Powell Daily Inflow and Release. Total volumes in legend are water year to date.

Data wrangling strategy

1. Import the reservoir elevation-volume curves from the folder AutoReadUSBRData.
2. Import Daily, Monthly, and Annual Lake Powell and Lake Mead data from the Reclamation's HydroData Web portal https://www.usbr.gov/uc/water/hydrodata/reservoir_data/site_map.html. Or use saved data in the AutoReadUSBR folder.
3. Import the critical reservoir elevations/volumes from AutoReadUSBRData folder.
4. Manipulate those critical volumes to volumes to show on right axes of plots.
5. Filter the reservoir data for the appropriate variables/fields.
6. Add Minimum Drawdown and Catastrophic Elevations.

This is a beginning R-programming effort! There could be lurking bugs or basic coding errors that I am not even aware of. Please report bugs/feedback to me (contact info below)

Requested Citation

David E. Rosenberg (2026), "Why communicating minimum drawdown elevations for Lake Powell and Lake Mead can mitigate risks of volatile Colorado River floa." Utah State University. Logan, Utah. https://github.com/ImmersiveModelsColoradoRiver/MinimumDrawdownElevations.

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Table 1: Current Storage and Elevations

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ResName	Storage	Pool Elevation
Lake Powell	6.19	3535.89
Lake Mead	6.93	1039.29

Figure 1: Lake Powell Storage and Elevation over Time in relation to Consequences.

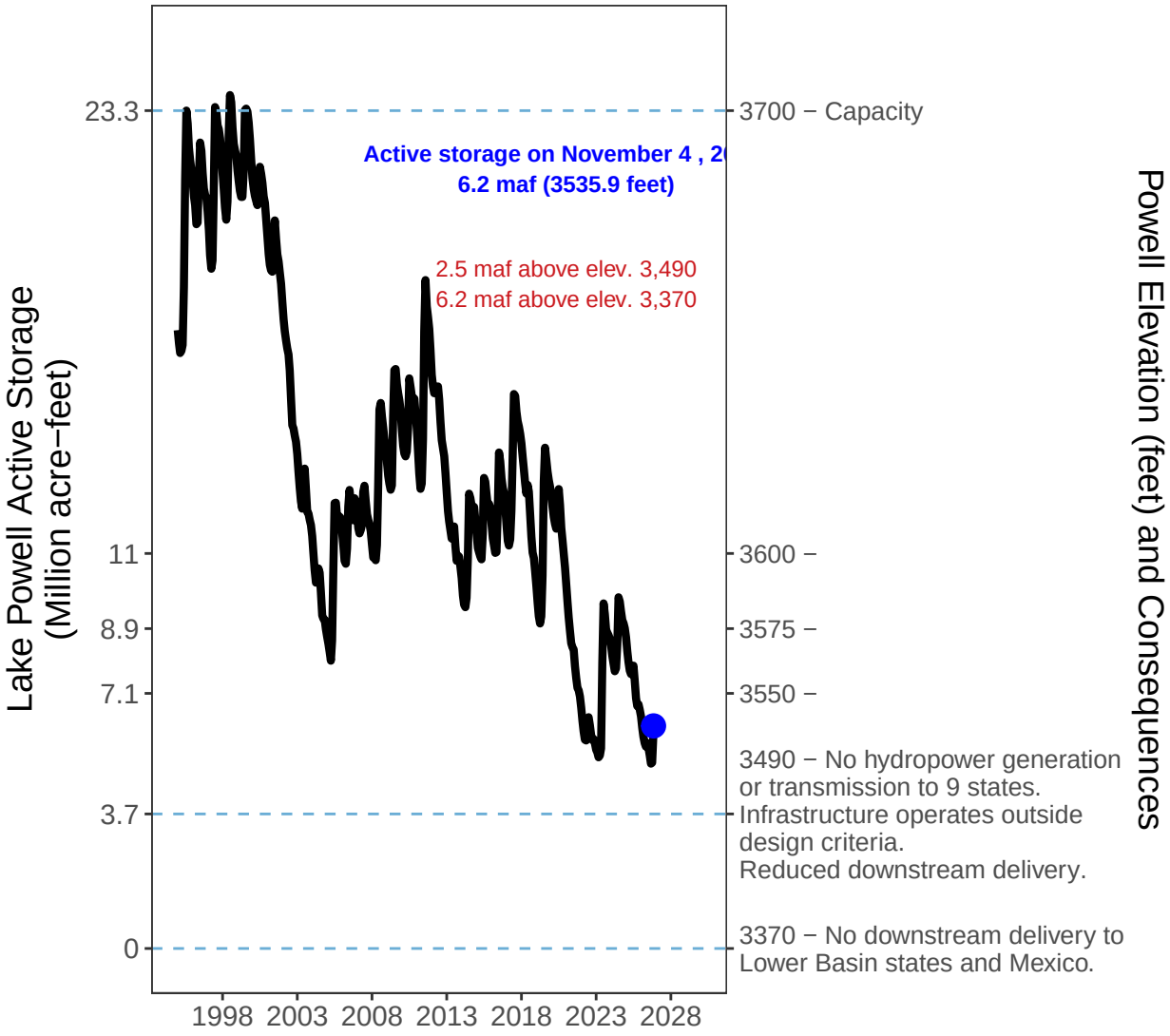


Figure 2: Lake Mead Storage and Elevation over Time and Consequences.

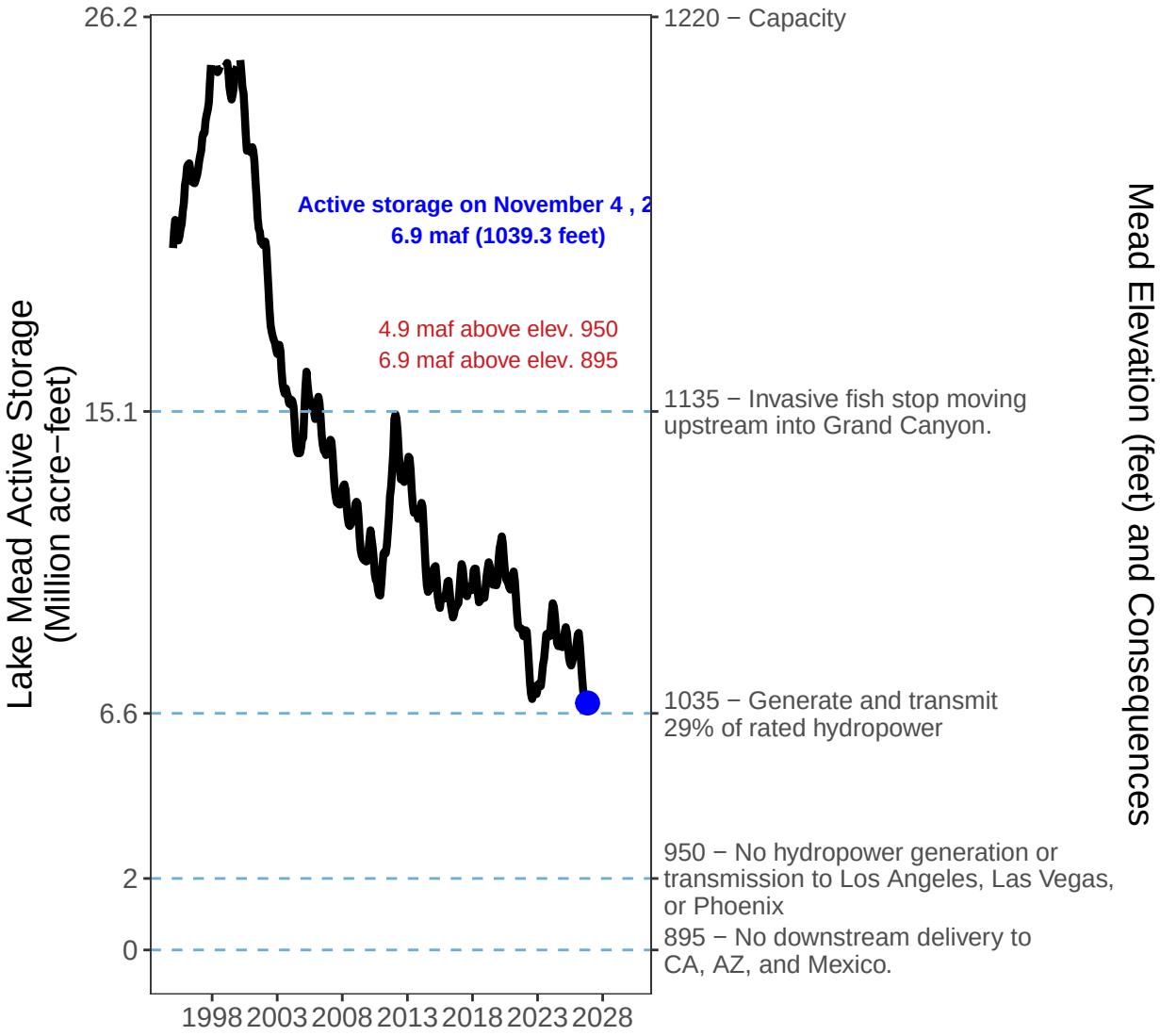


Figure 4: Lake Powell Minimum Drawdown and Catastrophic Elevations - Daily Data

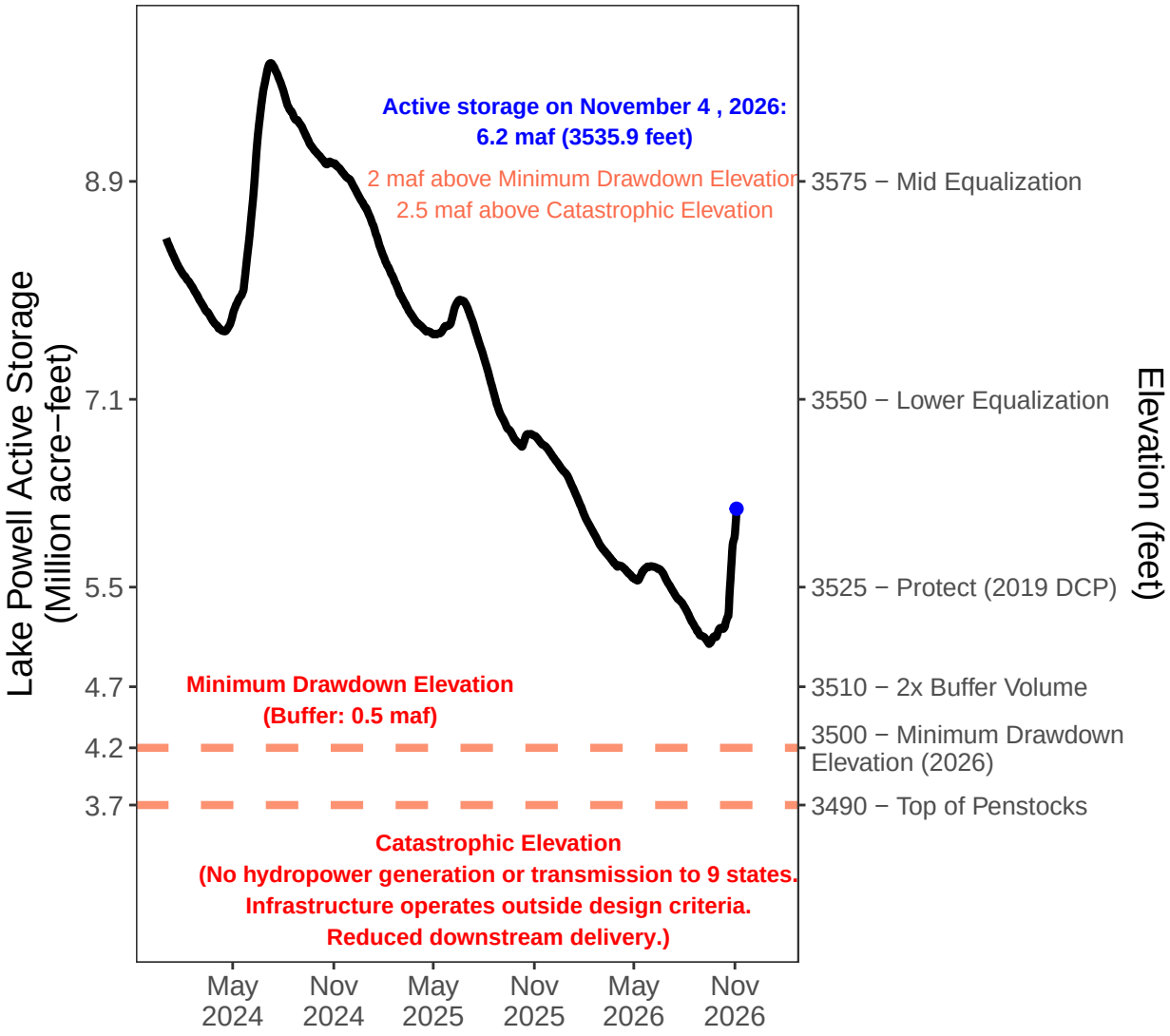


Figure AA: Lake Powell, Lake Mead, and Combined Storage

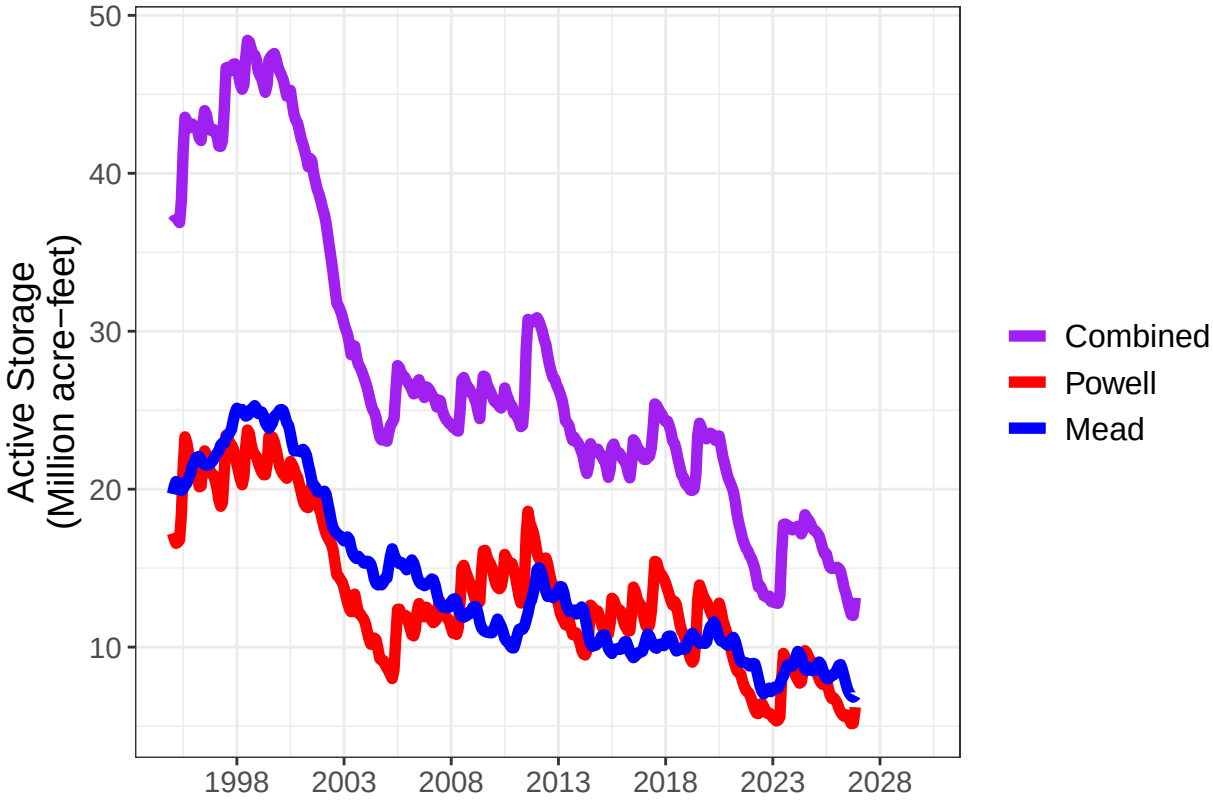


Figure BB: Lake Powell Daily Inflow and Release.

Total volumes in legend are water year to date.

